Ground and contact shadows: before / after (revised)

Lane VF-C, plan VF stages VF3 + VF4 · 2026-11-12 · branch vf-c-ground, after the review, build 1791

Each page

The same tank, seed, camera and light three times, left to right:

none = today's look

ground (VF3) = faint broad colour patches in the sand, dark flecks gathered into clumps (denser on slopes), a few coarse grains at the foot of the hardscape (freshwater sand only), the cross-section through the front glass denser and darker toward the bottom

contact + ground (VF4) = plus soft shadows on the ground under wood, rock and plant clumps, and the lowest part of each rock and piece of wood a little darker where it meets the ground

Changes since the first PDF

same front camera in every column (script bug: a close-up changed the saved camera); small low pebbles get a small faint shadow; no pale grains on reef sand; clumps no longer depend on where the hardscape is (they stayed behind when a piece was dragged)

Cost

Measured on alternating frames, contact objects shown vs hidden (1050 x 850, Intel Arc Pro):

Tank	GPU ms before → after	Paired change	Draw calls
Planted 4 ft (79 gal)	28.99 → 29.39	+0.17 ms	406 → 409
Nano (10.7 gal)	10.59 → 10.58	0.00 ms	231 → 232
Reef bow-front	20.82 → 20.80	+0.02 ms	351 → 352

The machine was busy (absolute ms high; p95 spread up to 1.7 ms): the paired change is within noise. Both effects now on by default.

Planted 4 ft: front view



none | **ground (VF3)** | **contact + ground (VF4)**

Planted 4 ft: substrate close-up



none | **ground (VF3)** | **contact + ground (VF4)**

Planted 4 ft: largest wood / rock close-up



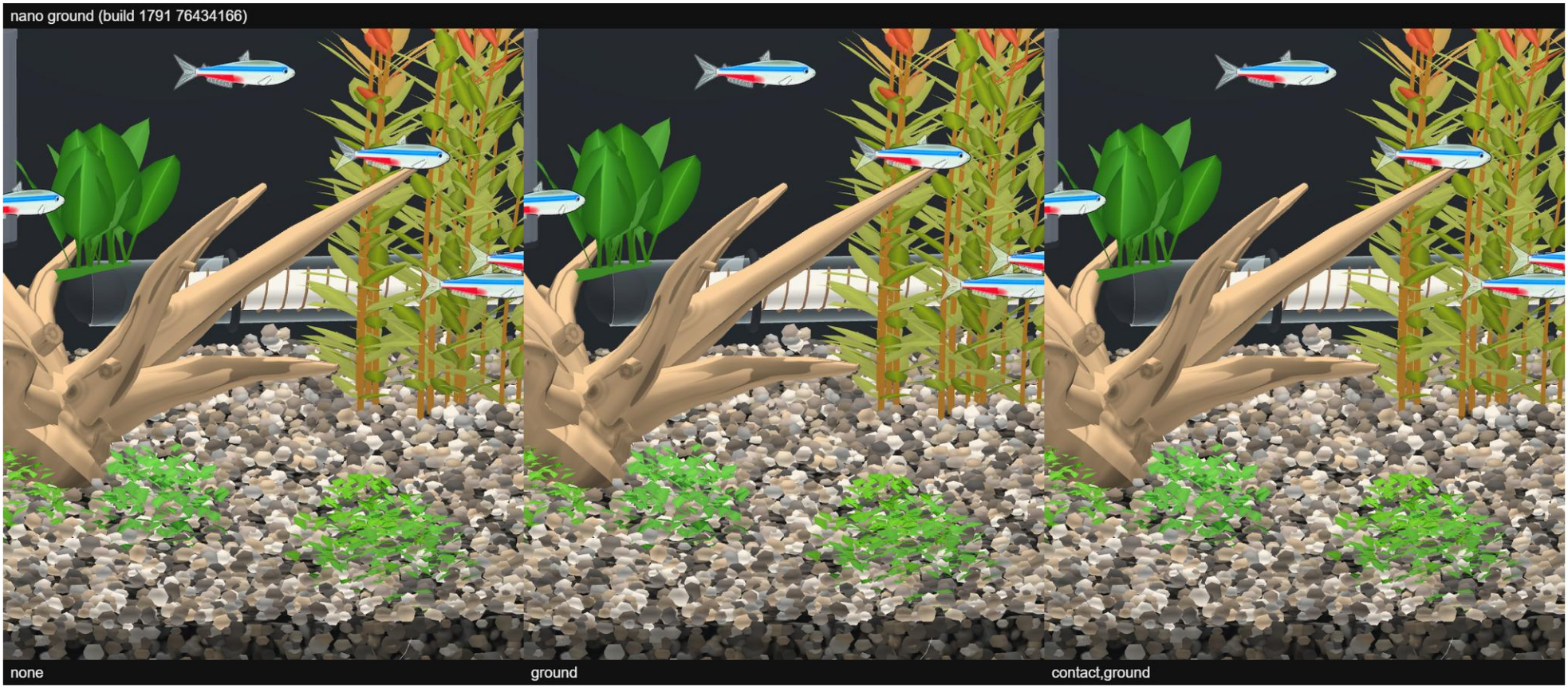
none | **ground (VF3)** | **contact + ground (VF4)**

Nano rimless: front view



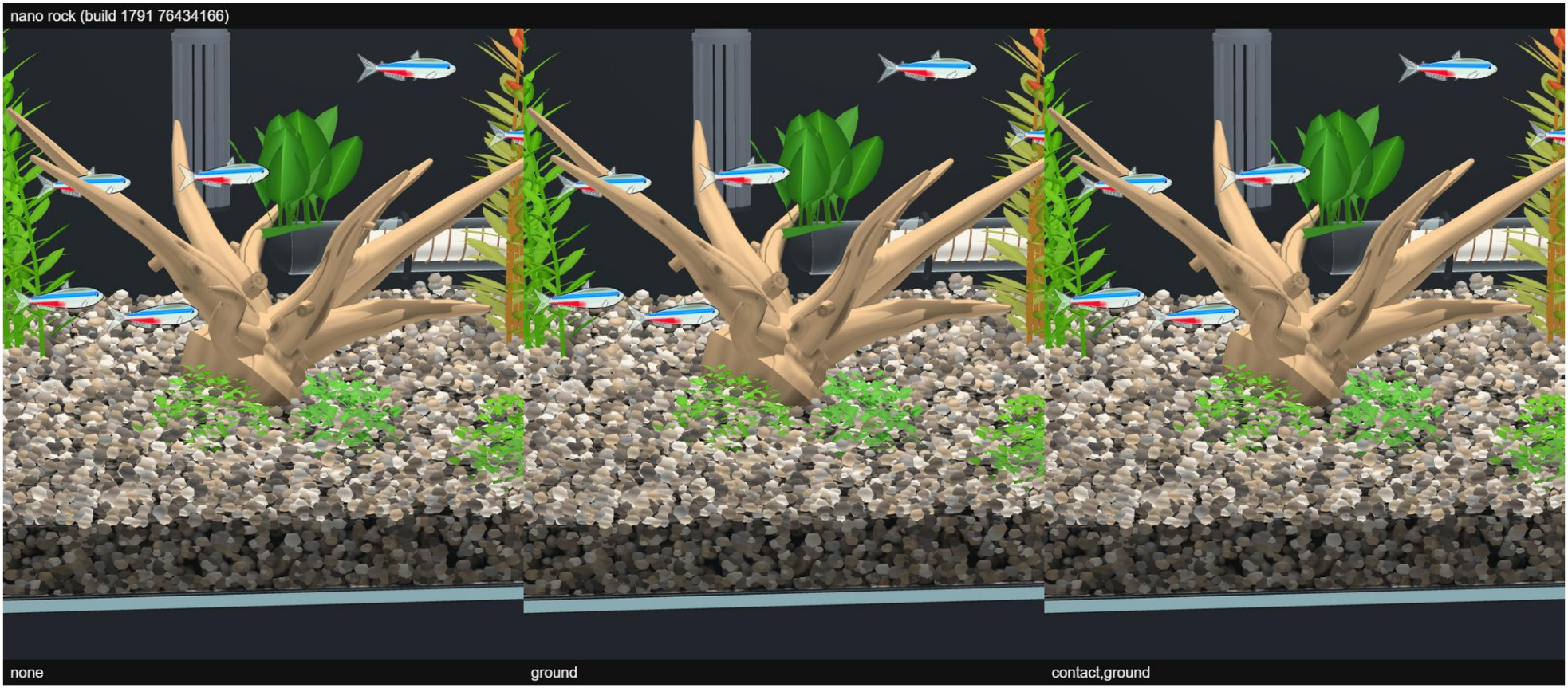
none | **ground (VF3)** | **contact + ground (VF4)**

Nano rimless: substrate close-up



none | **ground (VF3)** | **contact + ground (VF4)**

Nano rimless: largest wood / rock close-up



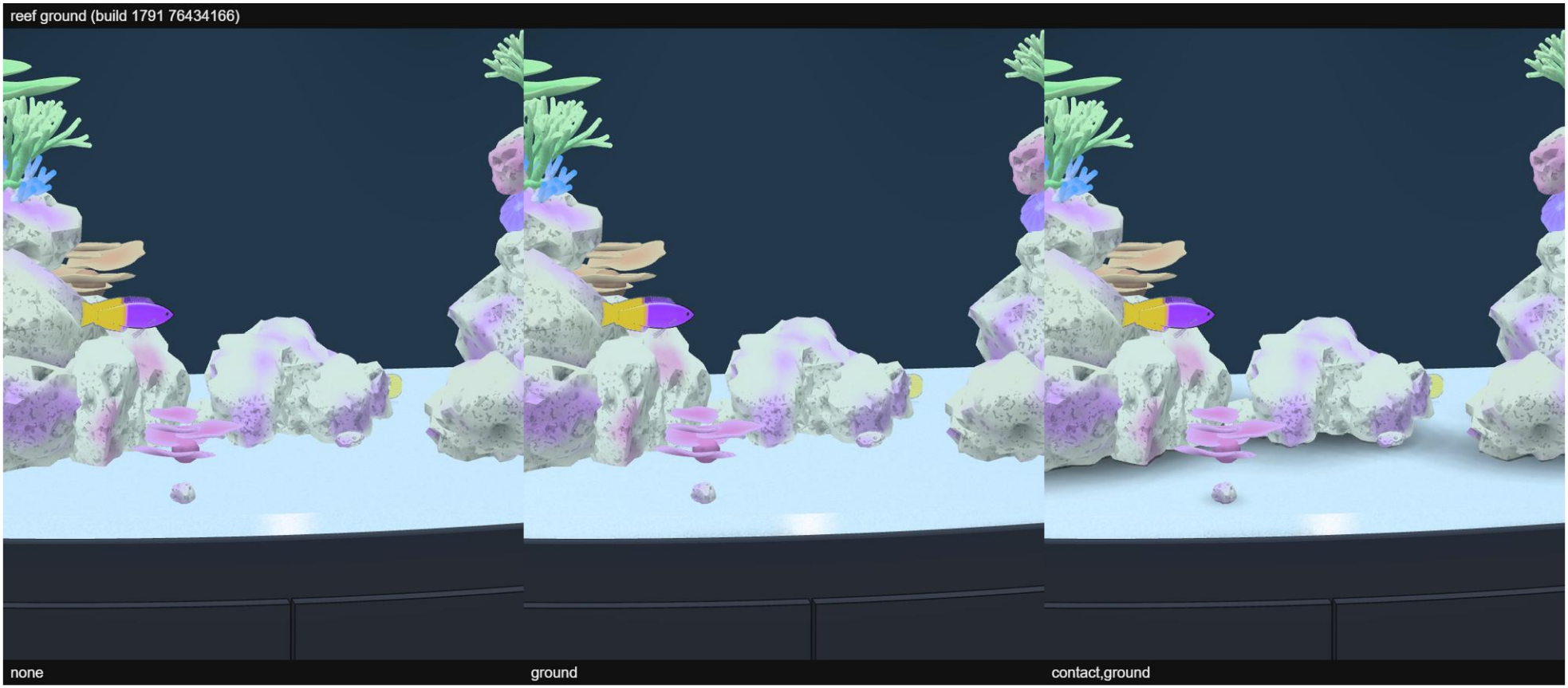
none | **ground (VF3)** | **contact + ground (VF4)**

Reef bow-front: front view



none | **ground (VF3)** | **contact + ground (VF4)**

Reef bow-front: substrate close-up



none | **ground (VF3)** | **contact + ground (VF4)**

Reef bow-front: largest wood / rock close-up



none | **ground (VF3)** | **contact + ground (VF4)**